PRATHAM P PANSARE

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ABOUT ME

Dedicated Software Developer with hands-on experience in building scalable, full-stack applications and AI-driven platforms. Proficient in modern web frameworks and relational databases, with a strong ability to translate complex technical requirements into user-centric solutions. Passionate about software architecture, clean code, and continuous learning.

EDUCATION

A. P. Shah Institute of Technology (Engineering), Thane

B.Tech in Information Technology

Jana Gana Mana College, Dombivli

Higher Secondary Education (11th - 12th)

Model English School, Dombivli West

Secondary Education (1st - 10th)

SKILLS

Languages: Python, JavaScript, CSS, HTML, SQL

Libraries/Frameworks: React, Next.js, Flask, Tailwind CSS, Bootstrap

Databases: MySQL, PostgreSQL, SQLite, Supabase

Cloud & DevOps: AWS, GitHub Actions, Vercel

Tools: Figma, Canva, Google Colab, GitHub

PROJECTS

ShetVaidya (Plant Disease Analyzer) — **React, FastAPI, PostgreSQL, TensorFlow**

- Built a production-grade, location-aware AI crop disease analyzer and medicine verification platform aimed at empowering farmers with immediate diagnostic tools.
- Developed an accurate machine learning inference microservice utilizing a TensorFlow MobileNetV2 model to analyze uploaded plant images and identify specific diseases.
- Engineered a robust and scalable backend using FastAPI, integrating JWT-based HttpOnly cookies for secure authentication and Google OAuth for user login.
- Configured Cloudflare R2 object storage to securely handle and validate image uploads, enforcing strict 5MB size limits and MIME type verifications.
- Built a responsive frontend application with React and Vite, featuring protected routes and a ZXing barcode scanner for medicine batch verification.
- Established a complete CI/CD pipeline using GitHub Actions to automate static checks, builds, and containerized deployment across Vercel and Hugging Face Spaces.
- Integrated PostgreSQL for structured relational data and Redis for caching, ensuring high-performance data retrieval and session management.

Khatawali Records

React, Supabase, Bootstrap

- Engineered a mobile-first, comprehensive credit and debit ledger application designed for small business owners and individuals to manage financial transactions.
- Packaged the application natively for Android using Capacitor, enabling seamless mobile features like an in-app contact importer and deep linking capabilities.
- Designed a scalable PostgreSQL database schema on Supabase, leveraging Row Level Security (RLS) policies and Supabase Auth to guarantee user data privacy.
- Implemented robust category-wise separation, allowing users to track the same person across multiple independent ledger categories seamlessly.
- Developed an advanced filtering and search dashboard supporting date-range queries, year/month summaries, and instant person-lookup by name or phone number.
- Integrated high-value export functionalities utilizing jsPDF and PapaParse, allowing users to download per-person transaction histories as PDFs or Excel files.
- Enhanced application security and user experience by implementing biometric fingerprint locking mechanisms configurable directly within the app settings.

Scholar Nexus

Next.js, TypeScript, Tailwind CSS, Prisma

- Architected an AI-powered, multi-source research paper discovery and synthesis platform that parallel queries nine major academic databases simultaneously.
- Developed an advanced deduplication algorithm that merges records by DOI and title similarity, ensuring the richest metadata is preserved across all sources.
- Implemented an intelligent ranking engine that scores papers across six dimensions including semantic relevance, citation impact, recency, and open-access availability.
- Integrated a cutting-edge AI evidence synthesis feature that analyzes multiple papers to identify consensus, contradictions, research gaps, and key findings.
- Built an interactive, D3.js force-directed citation network graph, allowing users to visually explore references and citing papers for any given publication.
- Embedded a PDF full-text Q&A chatbot capable of answering complex user questions by directly analyzing the extracted text of the research papers.
- Engineered personalized features including color-coded reading collections, automated daily/weekly search alerts, author profiles, and persistent search history.

Air Pollution Tracker V2.0

React, Tailwind CSS, Supabase, Mapbox

- Created a modern, data-driven React application that visualizes live and historical air-quality metrics for a catalog of over 200 cities worldwide.
- Implemented differentiated, role-based dashboards catering to both regular citizens (quick stats, advisories) and government officials (incident desks, heatmaps).
- Built interactive, high-density map views utilizing React Leaflet and Mapbox, featuring clustered markers and geolocation to seamlessly track air quality hotspots.
- Designed a robust data layer using React Query for stale-while-revalidate caching, optimistic updates, and background refreshing of OpenAQ API snapshots.
- Developed comprehensive Phase 5 analytics dashboards to deliver single-city deep dives, forecasting bands, health advisories, and source attribution analysis.
- Integrated Supabase for secure authentication, persistent offline-friendly user preferences, and an audit logging pipeline for monitoring government routing access.
- Optimized frontend performance with virtualized list rendering via react-window, ensuring smooth scrolling through hundreds of live environmental sensors.